Advanced VLSI Design - Final Project

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November 2026

1 Introduction

This report demonstrates the schematic design, simulation, and layout of an instruction register, 16-state counter, and ring counter in Electic VLSI. Each of these components uses pre-built components found in the *muddLib07.jelib* [3] and serve as key elements in a very simple microprocessor [5]. Transistor models were pre-designed using the *cmos_models.txt* [4] library. Simulations are tested through LTspice with validation through logisim simulations [2]. Successful completion of each schematic and layout is addressed in each section with unresolved layout errors additionally being addressed.

2 Instruction Register

The instruction register utilized eight flip-flops (flopr_c_1x), four tri-state buffers (buftri_c_1x), two inverters (inv_1x), and one and gate (and2_1x). The latch pin (LatchInstrReg) and main clock (MainClock) pins were connected to the and gate to produce a gated clock signal for ph1 of the flip-flops, while the gated clock was fed through an inverter into ph2 of the flip-flops. The clear signal was sent through an inverter to the reset pins of each flip-flop. The first four flip-flops in the instruction register were assigned Instr[3:0] at d and ToInstr[3:0] at q. The second set of four flip flops were assigned Data[3:0] at d and q outputs (q[3:0]) that are fed to the tri-state buffers. An enable signal was applied to all tri-state buffers and high-impedence buffer output pins were assigned (IB[3:0]).

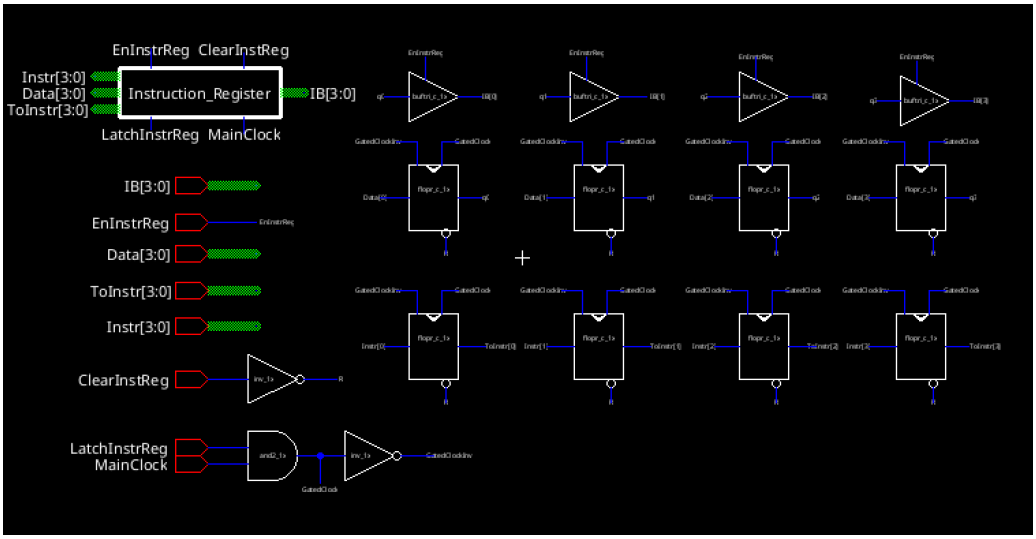


Figure 1: Instruction Register Schematic with Corresponding Icon View

To ensure proper functionality, the circuit was simulated by injecting a pulse signal into the main clock (MainClock), followed by piecewise functions into the clear (ClearInstReg), latch (LatchInstrReg), enable (EnInstrReg). The various data and instruction pins were given constant voltages of either 0 or 5V.

```
Vdd vdd 0 5
Vgnd gnd 0 0
Vclk MainClock 0 PULSE(0 5 0 0.1n 0.1n 5n 10n)
Vclr ClearInstReg 0 PWL(0 5 2n5 2.1n 0 100n 0)
Vlatch LatchInstrReg 0 PWL(0 0 15n 0 15.1n 5 35n 5 35.1n 0 100n 0)
Ven EnInstrReg 0 PWL(0 0 45n 0 45.1n 5 65n 5 65.1n 0 100n 0)
VD0 Data[0] 0 5
VD1 Data[1] 0 5
VD2 Data[2] 0 0
VD3 Data[3] 0 0
VI0 Instr [0] 0 0
VI1 Instr [1] 0 5
VI2 Instr [2] 0 0
VI3 Instr [3] 0 5
.tran 100n

.include *cmos_models.txt
```

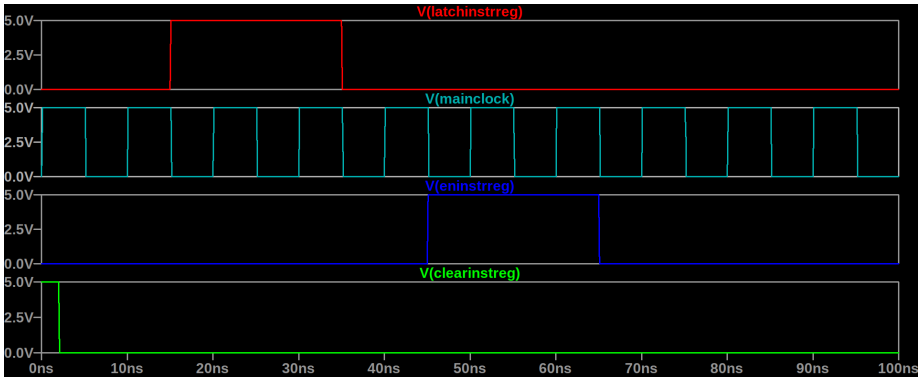


Figure 2: LTspice simulation results of clock and counter pins.

Figure 2 demonstrates the timing sequence of the four primary control signals driving the instruction register. The system is primarily driven by a main clock square wave of 10ns period, 50% duty cycle, and 5V logic-high amplitude. The signal is initiated by a 2ns active-high clear pulse to reset the internal flip-flops. After initialization, the data latch and output enable signals are spread sequentially as 20 ns, 5V pulses. To prevent overlap between these signals, they are separated by a 10ns delay, or one full clock cycle.

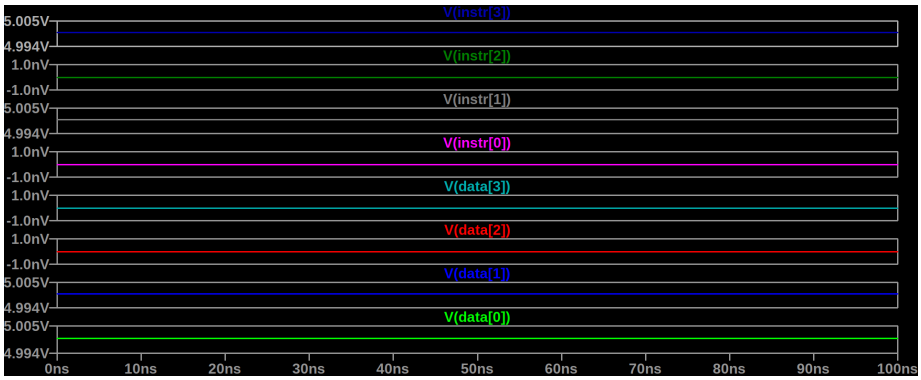


Figure 3: LTspice simulation results of data and instruction pins.

Figure 3 demonstrates the static DC logic levels applied to the two 4-input data buses during simulation. The register input is tied to a binary state of 1010, while the data bus is tied to a binary state of 0011 in order to provide known data values to verify latching and buffering operations.

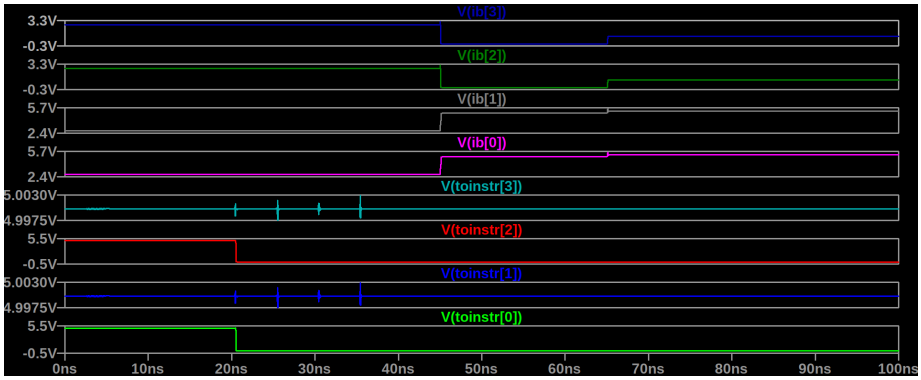


Figure 4: LTspice simulation results of toinstr and buffer output pins.

Figure 4 shows the response of the two 4-bit output buses resulting from the applied control signals and data inputs. The lower flip-flop register controls ToInstr and remains at a logic level zero until the latch signal enables the flip-flops. The ToInstr outputs then transition to reflect the 1010 state of Instr input bus. Some voltage spikes are present and representative of clock feed-through being visible on the scaled plots.

The layout of the instruction register was done sequentially in the following order (left-to-right): and gate, 2x inverters, 8x flip-flops, 4x tri-state buffers. The wiring followed the convention of the schematic seen in *Figure 1.* The gated clock and inverted gated clock were tied to ph1 and ph2 of each flip-flop using metal-2 and metal-1 connections running over the top of the layout. Similarly was done for the clear line. The second set of flip-flops were tied to their respective tri-state buffers through metal-2 and metal-1 connections running along the bottom of the layout. The greatest challenge when producing this circuit was ensuring minimal errors occurred; however, the end result of this layout saw 175 unresolved errors.

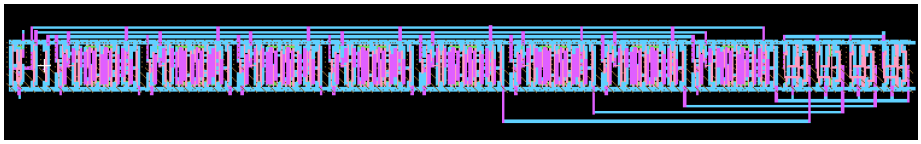


Figure 5: Instruction Register of Layout

3 Counter 16

The counter 16 circuit schematic utilized one and gate (and2_1x), six inverters (inv_1x), and four flip-flops (flopr_c_1x). The reason for needing 6 inverters is due to the fact each inverted q output of each flip-flop is fed back into its own d pin, and since the (flopr_c_1x) schematic from the *muddlib07.jelib* does not contain inverted pins for q, the schematic required making these connections. The main clock and enable signals were fed through an and gate and or gate to produce the standard and inverted gated clock signals. The only flip-flop requiring this signals is flip-flop zero as the rest of the flip-flop clocks are tied to the inverted and non-inverted q output.

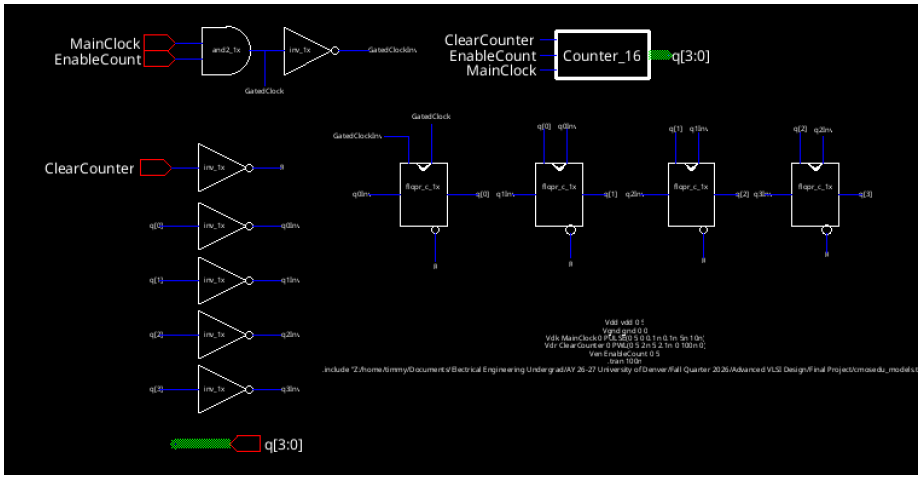


Figure 6: Counter 16 Schematic with Corresponding Icon

To ensure properly functionality of the counter, the circuit was simulated by injecting a pulse signal into the main clock (MainClock), followed by a piecewise function in the clear pin (ClearCounter) and the enable pin (EnableCount) set to 5V.

```
Vdd vdd 0 5
Vgnd gnd 0 0
Vclk MainClock 0 PULSE(0 5 0 0.1n 0.1n 5n 10n)
Vclr ClearCounter 0 PWL(0 5 2n 5 2.1n 0 100n 0)
Ven EnableCount 0 5
.tran 100n

.include cmos_models.txt
```

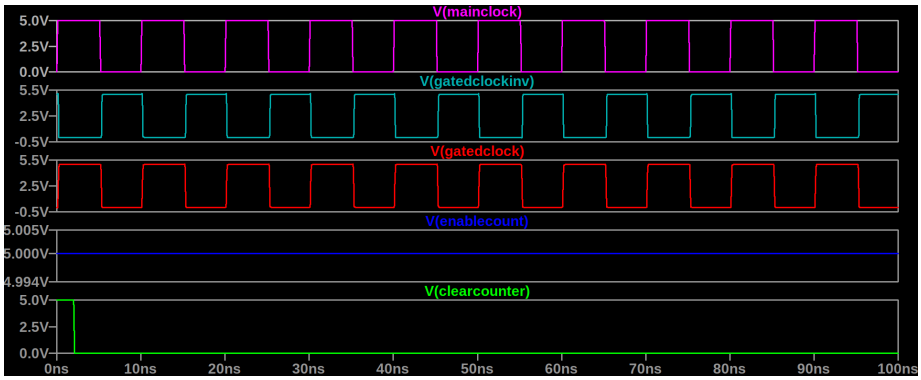


Figure 7: LTspice simulation results of the clock pins.

Figure 7 illustrates the control signals and clocks used to drive the 16 state counter. The signal

is initiated by a clear counter pulse that is pulled to ground to reset all the sequential flip-flops and allowing the counter to run. The counter is held at a constant 5V throughout the entire simulation. The plot also demonstrates the successful implementation of the inverted gated clock signals used to drive the first flip-flop.

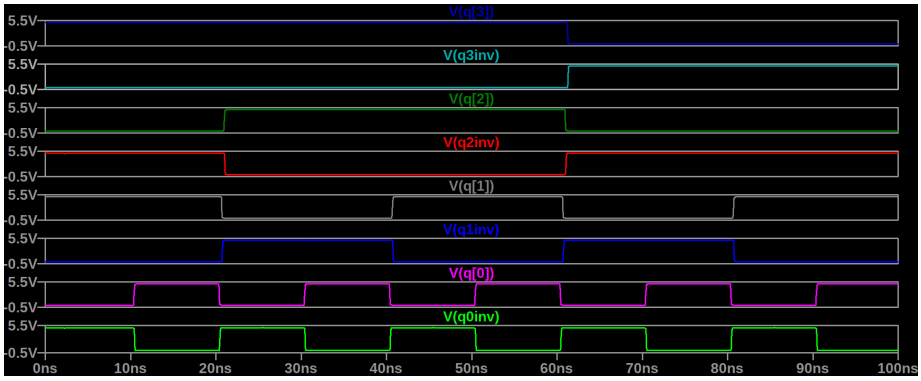


Figure 8: LTspice Simulation results of flip-flop inputs and outputs.

Figure 8 demonstrates the resulting output waveforms of the counter, displaying both q[3:0] and its inverted counterpart. Following the initial reset, the least significant bit (q[0]) toggles its state on every rising edge of the 10 ns clock in order to establish a base frequency. Because this counter ripples, each following flip-flop state toggles at half the frequency of the previous bit.

The layout for the 16-state counter utilized an alternating layout system that allowed output signals to remain close to gates. The order is as follows: and, inverter, inverter, flip-flop bit 0, inverter, flip-flop bit 1, inverter, flip-flop bit 2, inverter, flip-flop bit 3, inverter. The inverted clock cycle and reset pin were run along the top of the layout while the non-inverted clock cycle and signal-to-inverter connections were run along the bottom of the layout. This layout was the only one of all three to have no associated errors.

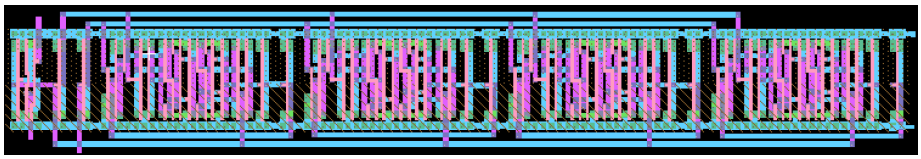


Figure 9: 16-state Counter Layout

4 Ring Counter 4

The ring counter 4 utilized four flip-flops (flopr_c_1x), four inverters (inv_1x), and four exclusive or gates (xor2_1x). Four inverters were needed to produce clear and phase counter signals, as well as an inverted q signal for bit 0 and 3. The clear pin and inverted and non-inverted phase signals were tied to each flip-flop. The flip-flops utilized a cascaded q configuration where each d pin was

controlled by the following flip-flops q output, with the exception of d[0], which was driven by q3inv. A similar procedure was performed on each exclusive or gate, with the only exception being q0inv driving xor0 along with q3. The outputs of the exclusive or gates were bused as Phase[3:0]

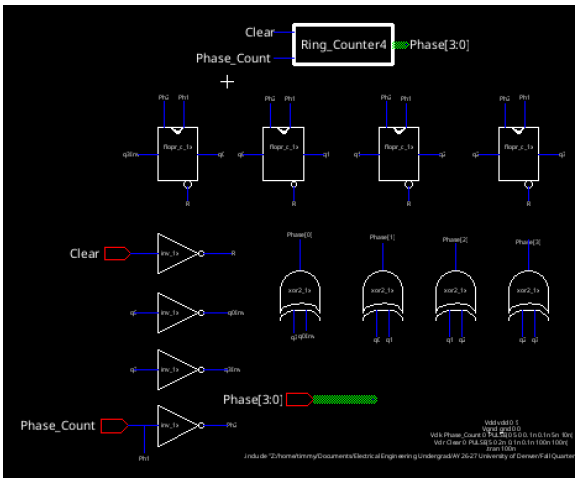


Figure 10: Ring Counter Schematic with Corresponding Icon

To ensure properly functionality of the counter, the circuit was simulated by injecting a pulse signal into the phase count (Phase_Count) and clear pin (Clear).

```
Vdd vdd 0 5
Vgnd gnd 0 0
Vclk Phase_Count 0 PULSE(0 5 0 0.1n 0.1n 5n 10n)
Vclr Clear 0 PULSE(5 0 2n 0.1n 0.1n 100n 100n)
.tran 100n

.include *cmos_models.txt
```

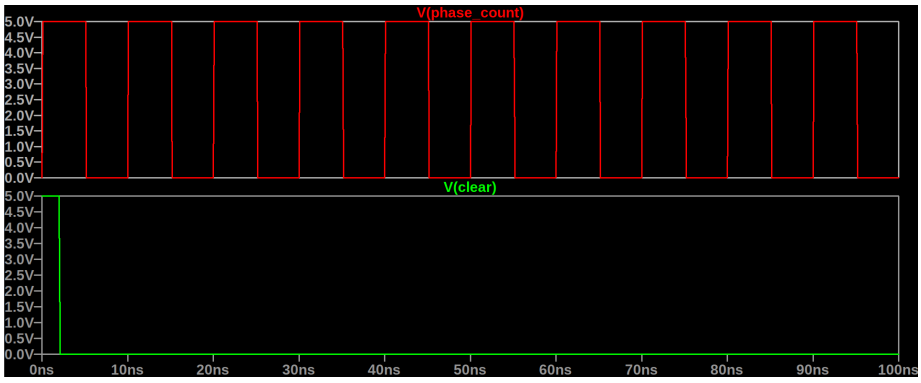


Figure 11: LTspice simulation results of the clear and phase counter.

Figure 11 illustrates the phase counter and clear signal used to drive the circuit. The phase counter is a continuous 10 ns square wave with a 5V amplitude, while the clear pin is pulled down to ground to reset the counters.

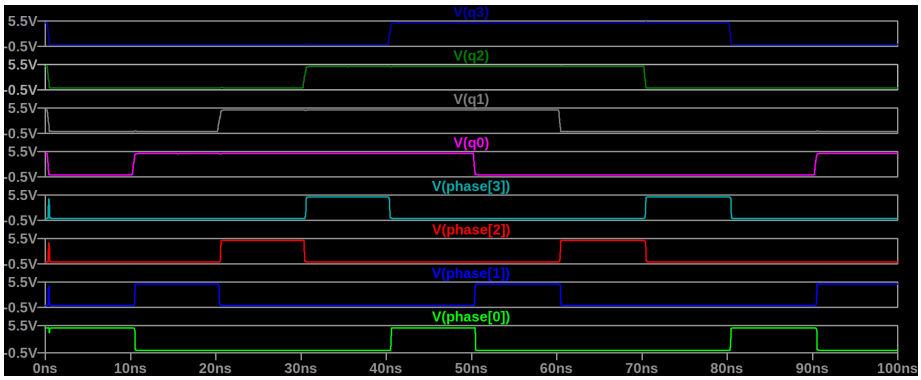


Figure 12: LTspice simulation results of phase and output pins.

Figure 12 demonstrates the output sequencing of the ring counter, displaying both the internal flip-flop states (q[3:0]), and the multiphase outputs from the exclusive or gates (Phase[3:0]). A single logic high state is shifted sequentially over q[3:0], creating a cascading overlap. The phase outputs transition sequentially, with only one phase active at a given time and remains present for one full clock cycle of 10ns.

The layout for the ring counter utilized an alternating system as opposed to a sequential system in order to maintain closed signal proximity between a flip-flop and its respective exclusive or gate. The order is as follows: inverter, inverter, flip-flop bit 0, inverter, exclusive or, flip-flop bit 1, exclusive or, flip-flop bit 2, exclusive or, exclusive or, flip-flop bit 3, inverter, exclusive or. The layout intended to minimize errors as much as possible, but resulted in 80 unresolved errors.

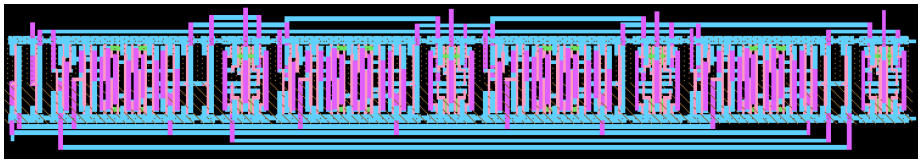


Figure 13: Ring Counter layout

5 Conclusion

This lab demonstrated the implementation of an instruction register, 16-state counter, and ring counter through Electric VLSI, intended for a very simple microprocessor circuit. A schematic, simulation, and layout was completed for each stage of the microprocessor. Simulations of the schematic demonstrated successful functionality of all three pieces, with all simulations being compared against logisim simulations provided by Dr. Martins [2]. The primary challenge within this

project was producing a layout with minimal errors, with most errors coming mostly from connecting ground and power rails of each component. This issue was unresolvable as the software would not flag this as an issue up until a third or fourth flip-flop was added to the layout. The only layout to be finalized with zero errors was the 16-state counter.

6 Documentation

[**1**] 2_Final_Project.pdf
[**2**] Very Simple Microprocessor Logisim schematics
[**3**] muddLib07.jelib
[**4**] cmosedu_models.txt
[**5**] Very Simple Microprocessor book, Chapter 3
[**6**] Google Gemini Pro AI for making code to simulate the schematics on LTspice.